

Project Executable Files

Date	7 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Executable Files

This section captures the executable and deployable files submitted for Smart Lender, and the instructions an evaluator needs to run the solution locally.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA — uses a flat CSV dataset (dataset/loan_approval_dataset.csv) and a serialized model (best_model.pkl); no relational database is used
5	Environment configuration file (.env.example or similar)	NA — no external secrets/API keys are used; note the Flask secret_key is hardcoded and should move to an env var before production
6	Deployed application URL (if hosted)	No — currently runs locally on http://localhost:5000; IBM Cloud Code Engine deployment planned
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web application only
8	Dockerfile / Containerization config (if applicable)	No — not yet containerized
9	Test files and test results	No — no automated test suite included yet
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

```
smart_lender/
├── app.py                # Flask application (routes, prediction logic)
├── train_pipeline.py     # Trains & compares models, saves best_model.pkl
├── best_model.pkl        # Serialized trained model, scaler, and encodings
├── requirements.txt      # Python dependencies
├── tailwind.config.js    # Tailwind CSS design tokens for the UI ── README.md
├── .gitignore
├── dataset/
│   └── loan_approval_dataset.csv
├── templates/
│   ├── index.html        # Landing page
│   └── loanEligibility.html # Applicant input form
├── static/
│   ├── images/
│   └── charts/           # Generated charts used by dashboard/eda pages
├── AnalyzePredictAnimation.html # Loading/analysis animation
├── prediction.html        # Prediction results page
├── dashboard.html        # Model performance dashboard
└── eda.html              # Exploratory data analysis view ──
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not yet deployed — runs locally at http://localhost:5000
Login Credentials (Demo)	Not applicable — no authentication/login is required to use the demo
Platform / Hosting Provider	IBM Cloud Code Engine (planned; not yet deployed)
Repository Link	https://github.com/nikhilhanumantu/Smart-Lender
Demo Video Link	https://drive.google.com/file/d/18ln4h7AFwKiw4q4MI5vW0qk2LC_HJrWO/view?usp=drive_link

Step 4: Run Instructions

Pre-requisites:

- Python 3.10 or higher
- pip (Python package manager)

Step 1: Clone or extract the project, then move into the folder: `cd smart_lender` Step 2:

(Recommended) Create and activate a virtual environment: `python -m venv venv &&`

`source venv/bin/activate` (Windows: `venv\Scripts\activate`)

Step 3: Install dependencies: `pip install -r requirements.txt`

Step 4: (Optional) Retrain the model if `best_model.pkl` is missing or outdated: `python train_pipeline.py`

Step 5: Start the application: `python app.py`

Step 6: Open the app in your browser at: `http://localhost:5000`

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	eda.html references additional univariate/bivariate/multivariate chart images not bundled in static/images/charts/	Regenerate the missing charts via an extended EDA script, or trim the template's image references
2	Flask <code>secret_key</code> in <code>app.py</code> is hardcoded	Move to an environment variable (e.g. <code>os.environ['SECRET_KEY']</code>) before any production deployment
3	No automated test suite included	To be added — recommend unit tests for the preprocessing pipeline and prediction route